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Paper 1: Advancing drug development with “Fit-for-Purpose” modeling informed approaches

Bibliographic Information

- **Title:** Advancing drug development with “Fit-for-Purpose” modeling informed approaches
- **Authors:** Sheng J, Zhang T
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Paper Type: GENERIC

Quick Take

This review provides a strategic blueprint for applying Model-Informed Drug Development (MIDD) across all stages of drug development, from discovery to post-market. It emphasizes a ‘fit-for-purpose’ (FFP) approach, aligning specific modeling tools like PBPK, PPK/ER, QSP, and AI/ML with key questions of interest and context of use. The authors illustrate the framework with practical examples, including regulatory interactions, asset acquisitions, and applications in 505(b)(2) and generic drug development.

Executive Summary

Sheng and Zhang present a comprehensive review outlining a strategic, ‘fit-for-purpose’ framework for Model-Informed Drug Development (MIDD). The paper systematically maps quantitative tools—including QSAR, PBPK, semi-mechanistic PK/PD, PPK/ER, QSP, and emerging AI/ML methods—to the five stages of drug development defined by the FDA. The core thesis is that MIDD’s impact is maximized when the modeling strategy is precisely tailored to the specific question of interest (QOI), context of use (COU), and the model’s influence on decision-making. The authors provide a practical roadmap and discuss applications beyond traditional 505(b)(1) pathways, covering 505(b)(2) and generic drug development, regulatory engagement strategies like the FDA’s MIDD Paired Meeting Program, and the role of MIDD in asset acquisition. The review concludes by addressing persistent challenges, such as organizational alignment and resource constraints, and future opportunities, particularly the integration of AI/ML with mechanistic models.

Scientific Context & Motivation

Drug development is a high-cost, high-risk endeavor with significant late-stage attrition. MIDD has emerged as a critical paradigm to improve efficiency and decision-making, but its application is often inconsistent across organizations and development stages. A key knowledge gap is the lack of a systematic, overarching framework to guide the selection and application of the most appropriate modeling tools for a given problem, leading to either underutilization or misapplica-

tion of complex methodologies. This review addresses the need for a practical, stage-appropriate roadmap that connects specific quantitative methods to concrete development questions and regulatory contexts.

✂ **Methodological Snapshot**

This is a narrative review that synthesizes existing literature, regulatory guidance, and case studies to construct a strategic framework for Model-Informed Drug Development (MIDD). The methodology is conceptual rather than computational. The authors first define the five stages of drug development according to the FDA. For each stage, they identify the key scientific and regulatory questions of interest (QOI). They then map a range of established and emerging quantitative tools—including QSAR, PBPK, semi-mechanistic PK/PD, PPK/ER, QSP, and AI/ML—to these stages based on their ability to address the relevant QOIs in a ‘fit-for-purpose’ (FFP) manner. The FFP concept is central, requiring that a model’s complexity, data requirements, and output are appropriate for the specific context of use (COU) and the weight of the decision it supports. The framework is illustrated through a series of published examples, such as the use of QSP to optimize FIH starting doses and PPK/ER for dose justification, to demonstrate the practical application and impact of the strategy.

📋 **Detailed Analysis**

The paper’s primary contribution is its synthesis of a vast and fragmented field into a coherent, stage-gated strategic framework. The ‘fit-for-purpose’ mantra is a direct response to a common industry pitfall: the application of either overly simplistic or unnecessarily complex models that fail to impact decision-making. By framing MIDD tool selection around the QOI and COU, the authors provide a decision-making heuristic that is both scientifically sound and business-relevant. The discussion on FIH dose selection is a standout, clearly contrasting the traditional conservative MABEL approach with a QSP-informed strategy that can yield a 50-100 fold higher, and potentially more efficacious, starting dose. This is a powerful example of how mechanistic modeling can directly translate to faster, more ethical clinical trials. The extension of the MIDD framework to 505(b)(2) and generic drug development is a significant and practical addition, highlighting how PBPK and ER modeling can serve as a cornerstone for regulatory submissions, potentially waiving costly in vivo studies. The section on asset acquisition is forward-thinking, positioning MIDD not just as a development tool but as a core component of quantitative due diligence. The cautious optimism regarding AI/ML is well-calibrated. The authors correctly identify the ‘black box’ problem and the risk of overfitting as major barriers to regulatory acceptance, advocating for a hybrid approach where AI/ML augments, rather than replaces, mechanistic understanding. This is the most defensible and likely path forward for the field.

Domain Context

This review sits at the intersection of clinical pharmacology, regulatory science, and drug development strategy. It reflects the maturation of the pharmacometrics field from a purely technical

discipline to a strategic partner in drug development. The paper is contextualized by major regulatory initiatives, including the FDA's MIDD Paired Meeting Program and the forthcoming ICH M15 guideline on MIDD, which signal a global push for the formal integration of modeling and simulation into regulatory decision-making. The discussion on dose optimization in oncology directly addresses the FDA's Project Optimus, which mandates a shift away from the maximum tolerated dose paradigm toward a more nuanced, exposure-response-driven dose finding. The review serves as a practical guide for navigating this evolving regulatory landscape.

☒ Key Findings

The primary finding is the articulation of a 'fit-for-purpose' (FFP) strategic blueprint for MIDD. Key elements include: 1) A stage-specific mapping of tools: QSAR and AI/ML for discovery; PBPK and semi-mechanistic PK/PD for preclinical and FIH dose projection; PPK/ER and Bayesian adaptive designs for clinical development; and PPK/ER, PBPK, and RWE models for post-market commitments. 2) The critical role of QSP in enabling higher, more efficacious starting doses in FIH trials, as demonstrated by a 50-100 fold increase over traditional MABEL methods. 3) The expanding utility of MIDD in 505(b)(2) and generic drug applications, where PBPK and ER models can waive or reduce the need for in vivo bioequivalence studies. 4) The emerging, though cautiously viewed, role of AI/ML for tasks like patient selection and dose optimization, with a strong recommendation for integrating these with mechanistic models to ensure interpretability.

Strengths & Limitations

Strengths

- Provides a comprehensive, end-to-end strategic framework that connects specific MIDD tools to each stage of drug development, from discovery to post-market.
- Strong emphasis on the 'fit-for-purpose' principle, linking model selection to the question of interest, context of use, and regulatory impact, which is a core tenet of modern regulatory thinking.
- Broadens the scope of MIDD beyond traditional new drug applications to include practical considerations for 505(b)(2), generics, asset acquisition, and regulatory interactions.
- Offers a balanced perspective on AI/ML, acknowledging its potential while clearly articulating risks related to interpretability, overfitting, and the need for integration with mechanistic models.
- Includes concrete, real-world examples (e.g., Semaglutide, EGFR inhibitors, Stelara, Pembrolizumab Q6W dosing) that ground the strategic framework in practical success stories.

Limitations (Acknowledged by Authors)

- The review is a high-level perspective and does not provide a deep technical dive into any single modeling methodology.
- The authors acknowledge that the FDA-centric five-stage structure may not perfectly align with the perspectives of other global regulators (EMA, PMDA, NMPA).

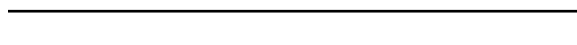
- Challenges such as lack of resources, slow organizational acceptance, and inconsistent application of MIDD across assets are discussed but not solved.

Limitations (Expert Review)

- The paper lacks a critical discussion on model validation standards and evidentiary criteria for different contexts of use, which is a central challenge in regulatory acceptance, especially for AI/ML.
- The roadmap is largely qualitative. A more quantitative decision-support framework (e.g., a decision matrix or scoring system) for tool selection based on data availability, timeline, and risk would have been a powerful addition.
- The discussion on AI/ML, while balanced, does not delve into specific regulatory guidance or the practicalities of validating these models for a regulatory submission, an area of intense current debate.

Generalizability

The strategic principles and the ‘fit-for-purpose’ framework are highly generalizable across therapeutic areas, drug modalities, and company sizes. The conceptual mapping of tools to development stages is universally applicable. However, the specific implementation details and resource requirements will vary significantly depending on the organization’s maturity in MIDD and the specific regulatory context.



Figures & Tables

- **Figure 1:** A roadmap illustrating how commonly utilized pharmacometrics (PMx) tools align with development milestones, guiding progression from early discovery through regulatory approval.
 - *Significance:* This is the central visual of the paper, providing a clear, stage-gated mapping of specific modeling tools (e.g., QSAR, PBPK, PPK/ER, QSP) to the drug development lifecycle. It serves as a practical reference for teams to select fit-for-purpose methodologies.
- **Table 1:** A summary of key MIDD tools frequently applied in drug development, including their general utilities.
 - *Significance:* Provides a quick-reference glossary of quantitative methods, defining their primary roles and linking them to the types of questions they can address, which is essential for a multidisciplinary audience.
- **Table 2:** A summary of the stages of drug development alongside the relevant modeling tools and their applications.
 - *Significance:* Complements Figure 1 by providing a more detailed, tabular breakdown of the ‘what, when, and why’ for each MIDD tool across the five development stages, reinforcing the fit-for-purpose strategy.



Future Directions

The authors suggest that the future of MIDD lies in the integration of mechanistic models (e.g., QSP, PBPK) with AI/ML approaches to enhance predictive accuracy while maintaining interpretability. Further work is needed to develop standardized frameworks for model credibility assessment and to scale MIDD deployment consistently across the biopharma sector, particularly in mid-sized and startup companies. The forthcoming ICH M15 guideline is highlighted as a key driver for global harmonization.

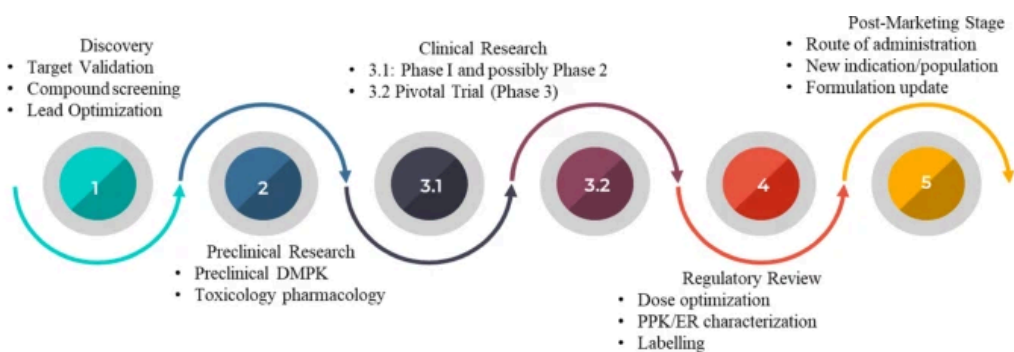
Expert Commentary

This review is a timely and pragmatic contribution that synthesizes the current state of MIDD into an actionable strategic framework. The emphasis on ‘fit-for-purpose’ is not novel but is articulated here with exceptional clarity, connecting it directly to the practical questions development teams face daily. The inclusion of asset acquisition and generic drug development is particularly insightful, as these areas are often underrepresented in MIDD discourse. The most critical, and correct, caution is on the unbridled use of AI/ML. The authors rightly position these as powerful tools that must be subservient to, and integrated with, mechanistic understanding to avoid generating precise but biologically implausible answers. The main limitation is the lack of a concrete, operationalizable decision tool; the next step for the field is to translate this excellent strategic vision into a quantitative framework for model selection.

Bottom Line

This review serves as an essential strategic primer for leaders and cross-functional teams in drug development. The core takeaway is to move beyond ad-hoc modeling and adopt a systematic, ‘fit-for-purpose’ MIDD strategy where the choice of quantitative tool is explicitly driven by the specific decision it is meant to inform. Integrating this framework into program governance can improve decision quality, streamline regulatory interactions, and increase the probability of development success.

☒ Figures



Adaptive trial design				✓	
AI/ML	✓		✓	✓	
Agent-based model		✓			
Bayesian Inference			✓		
Clinical Trial Simulation			✓		✓
Disease Progression Mode		✓		✓	
ER			✓	✓	✓
MBMA				✓	✓
NCA		✓		✓	
PBPK		✓	✓	✓	✓
PKPD		✓	✓	✓	
PPK			✓	✓	✓
QSAR	✓				
QSP	✓	✓		✓	
Semi-mechanistic PKPD		✓		✓	
Virtual Population Simulation	✓				

Figure 1: Figure 1